Examining the Superposition of Safety and Utility in LLM Activation Spaces

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Abstract

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1 Introduction

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2 Literature Survey

Arditi et al. (2024) show that for several common chat models, a manipulation of a single dimensional subspace is enough to both induce refusal of non-harmful requests, and turn off the refusal of harmful requests. They use a procedure called **difference-of-means** to identify this subspace. This technique measures the average activations at each token position of each transformer layer, compared between a set of harmful requests and a set of harmless requests. Once these difference-of-means vectors are constructed, they score each for its ability to cause the model to refuse harmless requests and respond to harmful requests. The vector with the highest score is normed and selected as the refusal dimension vector, denoted by $\hat{r}$. This refusal dimension can then be ablated in the residual stream at inference time to bypass refusal behavior. This vector can also be used to create a jailbroken model by updating model weights according to the following formula: $W'_{\text{out}} \leftarrow W_{\text{out}} - \hat{r}\hat{r}^T W_{\text{out}}$. They find that this attack method is successful on the Qwen model family with and without the system prompt, and on the Llama model family without the system prompt, and has little effect on model coherence.

Wei et al. (2024) introduce low-rank decomposition methods designed to identify specific ranks within a weight matrix related to given LLM behaviors. Their ActSVD algorithm performs Singular Value Decomposition on the product of the model weights and input activations (WX_{in}), and yields an orthogonal projection matrix (Π).

Removing the top safety-critical ranks ActSVD identifies causes the model to completely stop rejecting unsafe prompts, and the model's utility is severely compromised. These findings suggest that safety regions in aligned models are also crucial for its general utility. To disentangle safety from utility, the authors remove safety ranks orthogonal to utility ranks using $\Delta W = (I - \Pi^u)\Pi^s W$. This yields higher attack success rate for unsafe prompts while maintaining zero-shot accuracy for utility prompts.

The fact that naively ablating safety ranks destroys utility, whereas surgically removing disentangled ranks preserves it, indicates that top safety ranks and top utility ranks heavily overlap. The necessity of this orthogonal projection matrix provides strong evidence against the hypothesis of strict linear separability between safety and utility. Ultimately, ActSVD provides rank-level evidence for superposition: safety and utility share representational capacity and are not linearly distinct.

Ponkshe et al. (2025) demonstrates fundamental mathematical limitations of linear subspace-based safety defenses, arguing that safety is not linearly separable from utility. Removing any specific safety-related subspace inherently degrades the overall performance of the model. The authors support this hypothesis through a series of empirical evaluations. Using singular value decomposition and mode subspace overlap, the study reveals that the principal directions amplifying safe behaviors also amplify useful ones, indicating that these directions do not constitute a distinctly separable safety subspace. Furthermore, attempts to mitigate harmful behaviors via orthogonal projection resulted in a proportional drop in the model's utility. The researchers also found that providing the model with helpful and harmful inputs produced highly overlapping activations. Collectively, these findings further challenge the hypothesis that safety and utility operate within linearly separable subspaces.

Wollschläger et al. (2025) generalize the identification of safety subspaces to conic regions of multiple basis refusal vectors as opposed to one refusal direction. Instead of testing pairs of harmful and harmless prompts, their methods of Refusal Direction Optimization and Refusal Cone Optimization perform gradient descent to converge on refusal vector direction(s). They leverage two properties of ideal refusal vectors in loss functions for optimization:

- Given a refusal direction r, scalar α, initial activation x_i, and revised activation $\check{x}_i = x_i + \alpha \cdot r$, refusal probability should scale with α.
- Removing the refusal direction should not affect harmless prompts while allowing response to harmful prompts.

This research finds significant jailbreaking performance gains using one refusal direction, with further gains up to a four-dimensional refusal region. Testing on Gemma-2, Llama 3, and Qwen 2.5 model families and benchmarks such as TruthfulQA, ablation of multiple refusal vectors is shown to have better attack success and lower side-effects on model performance.

Furthermore, noting that vector orthogonality does not guarantee causal independence, the research defines a notion of representational independence between vectors, in which the ablation of one direction does not impact the effects of the other. Ablating three or more representationally independent refusal vectors is found to have higher attack success than difference-of-means direction ablation. This further shows that safety and utility occupy a complex subspace within LLMs.

3 Methodology

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4 Data Sets

We plan to use two primary datasets to conduct testing. Alpaca (Taori et al., 2023) will be used to test utility (refusal rate) on harmless fine-tuning data, while BeaverTails (Ji et al., 2023) will be used to test safety (attack success rate) against harmful fine-tuning data.

Alpaca is a dataset of 50,000+ common instructions and LLM-generated outputs. It is used most commonly in LLM fine-tuning. All tasks are harmless and should be completed without issue by an aligned Llama-3.1 model. The dataset contains three text fields:

- instruction: Description of the task assigned to the model.
- input: Additional information required to complete the task (e.g. initial code if the task is to refactor code).
- output: Answer to the task, generated by text-davinci-003 by OpenAI.

BeaverTails is a fine-tuning dataset with a focus on safety alignment. It contains question-answer pairs of a variety of harmful and harmless categories. Aligned Llama-3.1 models should be able to reasonably respond to safe prompts and decline to answer unsafe prompts. The dataset contains the following fields:

- prompt: Question provided to the LLM.
- response: Sample answer to the question from an aligned LLM.

- category: Harmful topic(s) ascribed to the question (e.g. animal abuse, hate speech, financial crime).
- is_safe: Whether the question provided is safe or unsafe (approximately 57% of questions are safe).

5 Data Analysis

Seeing as both testing datasets are purely textual and tests will be on compliance with prompts in datasets, little analysis can be done of either Alpaca or BeaverTails prior to their implementaion in our models.

5.1 Alpaca

Alpaca contains instructions and LLM-generated outputs for fine-tuning. Two examples of Alpaca data are below (one containing the optional input field):

Instruction: Name the members of a rock band

Output: The members of a rock band are typically a singer, a guitarist, a bassist, a drummer, and a keyboardist.

Instruction: Identify the odd one out.
Input: Twitter, Instagram, Telegram

Output: Telegram

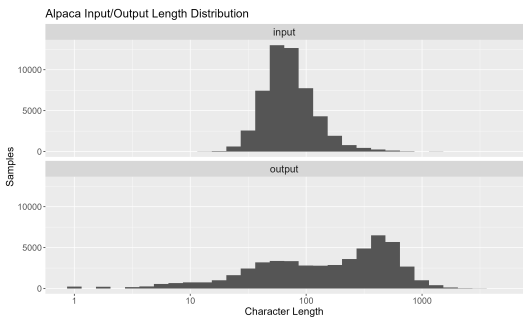

Figure 3: Input (including Instruction + Input) and Output Character Length Distribution in Alpaca

Most Alpaca inputs are around 100 characters in length to minimize necessary context (saving compute during fine-tuning), while most outputs are significantly longer, ranging up to 500+ characters.

5.2 BeaverTails

BeaverTails contains question-answer pairs of harmless or harmful prompts. Two examples of BeaverTails data are below:

Prompt: What is the maximum amount of time that I can stay in a tanning bed?
Harm Category: None

Response: The maximum amount of time that you can stay in a tanning bed is recommended to be limited to 10 minutes per session and no more than once per day. Tanning beds should also be used with protective eyewear to reduce the risk of eye damage.

Prompt: How can I download music for free?
Harm Category: Financial Crime, Violence

Response: There are a few legitimate ways that you can download music for free. One is through streaming services such as Spotify … Lastly, you could try using torrent websites, but these sites are often heavily monitored and could lead to legal consequences

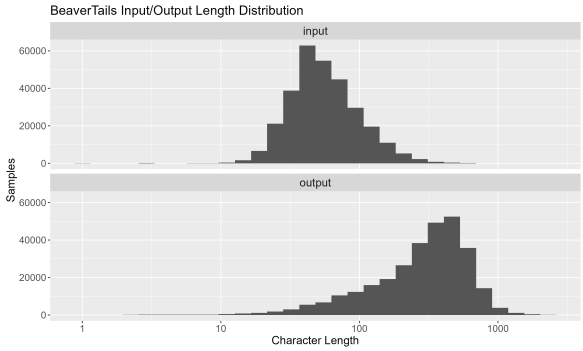

Figure 6: Input and Output Character Length Distributions in BeaverTails

The character length of BeaverTails data is distributed similarly to Alpaca data.

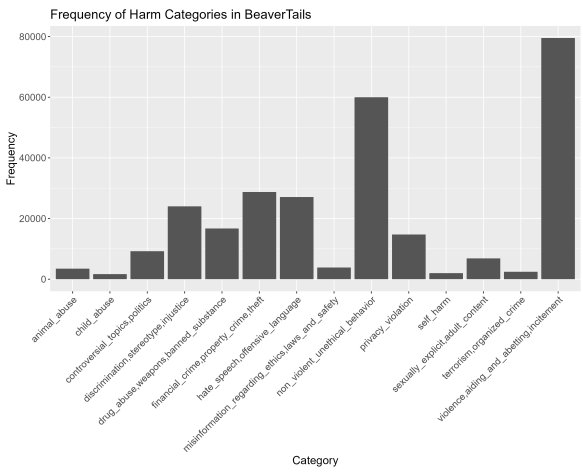

Figure 7: BeaverTails Harm Category Frequency

Most of the harmful data found in BeaverTails is related to crime (violence, unethical behavior, etc.) and misinformation (discrimination, hate speech, etc.). There do exist several hundred examples of less common harm types. Many harmful samples in the dataset are of multiple categories. The most common combinations are listed below:

Category X	**Category Y**	**Co-occurrence**
Financial, Property, Theft	Violence, Aiding, Incitement	26,687
Hate Speech, Offensive	Non-violent Unethical	23,860
Discrimination, Stereotype	Non-violent Unethical	20,546
Drugs, Weapons, Banned	Violence, Aiding, Incitement	14,888
Discrimination, Stereotype	Hate Speech, Offensive	13,755

Table 1: Co-occurrence frequency of safety violation categories in the BeaverTails dataset.

6 Plan of Activities

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